Shocks, Weak Solutions, and Entropy: An Exposition of Burgers' Equation

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1 Introduction

The viscous Burgers' equation

$$u_t + u\,u_x = \nu\,u_{xx}, \qquad \nu > 0, \tag{1}$$

was introduced by Burgers [2] as a one-dimensional model of turbulence; it had appeared earlier in work of Bateman [1]. It has the form of the Navier–Stokes momentum equation without the pressure term, and combines nonlinear advection with viscous diffusion. Setting $\nu = 0$ gives the inviscid equation

$$u_t + u\,u_x = 0, \tag{2}$$

the scalar conservation law $u_t + f(u)_x = 0$ with flux $f(u) = u^2/2$. Conservation laws of this type arise in gas dynamics [5, 10] and traffic flow [11].

We address the following question: *why does the inviscid Burgers' equation require an entropy condition, and why is that the correct condition?* Section 2 solves (2) by characteristics for C^1 data, and Section 3 computes the time at which classical solutions break down. Section 4 introduces weak solutions and the Rankine–Hugoniot condition, and Section 5 shows that weak solutions are not unique. Sections 6 and 7 introduce the Lax and Oleinik entropy conditions and solve the Riemann problem. Section 8 derives the Cole–Hopf formula, states (citing Hopf) that the entropy solution is the vanishing viscosity limit, and examines Godunov's method numerically. The theory follows [5, §3.4].

2 Method of Characteristics

Consider the initial value problem

$$\begin{cases} u_t + u\,u_x = 0, & x \in \mathbb{R},\ t > 0, \\ u(x,0) = u_0(x), & x \in \mathbb{R}, \end{cases} \tag{3}$$

with $u_0 \in C^1(\mathbb{R})$ in Sections 2 and 3. For discontinuous data, such as the Riemann data (19), (3) is understood in the weak sense of Definition 4.1.

2.1 Characteristics

A *characteristic* of a C^1 solution u is a curve $x(t)$ satisfying

$$\dot{x}(t) = u(x(t), t). \tag{4}$$

Along it,

$$\frac{d}{dt}\,u(x(t),t) = u_x\,\dot{x} + u_t = u_t + u\,u_x = 0. \tag{5}$$

Proposition 2.1 (Characteristics are straight lines)**.** *Let u be a C^1 solution of (3) on $\mathbb{R} \times [0,T)$, and let $x(t)$ solve (4) with $x(0) = x_0$. Then for $0 \le t < T$,*

$$u(x(t),t) = u_0(x_0), \qquad x(t) = x_0 + u_0(x_0)\,t. \tag{6}$$

Proof. By (5), $u(x(t),t) \equiv u_0(x_0)$, so (4) gives $\dot{x}(t) = u_0(x_0)$, which integrates to the stated formula. □

2.2 The implicit solution formula

Theorem 2.2 (Implicit solution via characteristics)**.** *Let $u_0 \in C^1(\mathbb{R})$, $\Phi_t(x_0) = x_0 + u_0(x_0)\,t$, and $T \in (0,\infty]$ such that for every $t \in [0,T)$, $\Phi_t' > 0$ on $\mathbb{R}$ and Φ_t maps $\mathbb{R}$ onto $\mathbb{R}$. Then the function u defined implicitly by*

$$u(x,t) = u_0(x_0), \qquad \text{where } x = x_0 + u_0(x_0)\,t, \tag{7}$$

is the unique C^1 solution of (3) on $\mathbb{R} \times [0,T)$.

Proof. Well-definedness and regularity. Since $\Phi_t' > 0$, Φ_t is strictly increasing, hence bijective by hypothesis, so $u(x,t) = u_0(\Phi_t^{-1}(x))$ is well defined. The function $F(x_0,x,t) = x_0 + u_0(x_0)t - x$ is C^1 with $\partial_{x_0}F = \Phi_t' > 0$, so by the implicit function theorem $x_0(x,t) = \Phi_t^{-1}(x)$ is C^1 in (x,t) jointly (it is locally the unique root, and by injectivity coincides with the global one). Hence $u = u_0(x_0(x,t))$ is C^1.

The PDE holds. Write $x_0 = x_0(x,t) = \Phi_t^{-1}(x)$. Differentiating $x = x_0 + u_0(x_0)\,t$ in x and in t gives

$$\frac{\partial x_0}{\partial x} = \frac{1}{1 + u_0'(x_0)\,t}, \qquad \frac{\partial x_0}{\partial t} = \frac{-u_0(x_0)}{1 + u_0'(x_0)\,t}, \tag{8}$$

both well defined since $\Phi_t'(x_0) = 1 + u_0'(x_0)t > 0$. By the chain rule applied to $u(x,t) = u_0(x_0(x,t))$,

$$u_x = \frac{u_0'(x_0)}{1 + u_0'(x_0)\,t}, \qquad u_t = \frac{-u_0(x_0)\,u_0'(x_0)}{1 + u_0'(x_0)\,t}, \tag{9}$$

so $u_t + u\,u_x = 0$, and $u(x,0) = u_0(x)$ since $\Phi_0 = \mathrm{id}$.

Uniqueness. Let v be any C^1 solution of (3) on $\mathbb{R} \times [0,T)$, and fix $t \in (0,T)$, $x \in \mathbb{R}$. Solve backwards in time the ODE $\dot\gamma(s) = v(\gamma(s),s)$, $\gamma(t) = x$ (the right-hand side is C^1, hence locally Lipschitz, so there is a unique maximal solution). By (5), v is constant along γ, hence so is $\dot\gamma$: γ is a straight line with bounded slope $v(x,t)$. It therefore stays bounded and cannot blow up, so it extends back to $s = 0$. Setting $x_0 = \gamma(0)$ we get $v(x,t) = v(x_0,0) = u_0(x_0)$ and $x = x_0 + u_0(x_0)t$. Thus every point (x,t) lies on a characteristic coming from $t = 0$, with $x_0 = \Phi_t^{-1}(x)$, and $v(x,t) = u_0(\Phi_t^{-1}(x)) = u(x,t)$. □

Corollary 2.3 (Local existence)**.** *Let $u_0 \in C^1(\mathbb{R})$ with $m := \inf_{x_0} u_0'(x_0) > -\infty$, and set $T = \infty$ if $m \geq 0$ and $T = -1/m$ if $m < 0$. Then (3) has a unique C^1 solution on $\mathbb{R} \times [0,T)$, given by (7).*

Proof. Fix $t \in [0,T)$ and put $\kappa_t := 1 + \min(m,0)\,t$, so $\kappa_t > 0$ and $\Phi_t'(x_0) = 1 + u_0'(x_0)t \geq \kappa_t$ for every x_0. Two consequences follow. (i) Φ_t is strictly increasing, hence injective. (ii) By the mean value theorem $\Phi_t(x_0) - \Phi_t(0) \geq \kappa_t x_0$ for $x_0 \geq 0$ and $\leq \kappa_t x_0$ for $x_0 \leq 0$, so $\Phi_t(x_0) \to \pm\infty$ as $x_0 \to \pm\infty$; by continuity Φ_t is onto $\mathbb{R}$. Hence the hypotheses of Theorem 2.2 hold on $[0,T)$. □

Remark 2.4. The uniform lower bound $\Phi_t' \geq \kappa_t > 0$, not strict monotonicity alone, gives surjectivity: for $u_0(x) = (\arctan x - x)/t_0$ one has $\Phi_{t_0}(x) = \arctan x$, strictly increasing with $\Phi_{t_0}' > 0$ but not onto $\mathbb{R}$. The hypothesis $m > -\infty$ cannot be dropped either: for $u_0(x) = x^2$, $\Phi_t(x) = x + tx^2$ is not injective for any $t > 0$, since $\Phi_t'(x) < 0$ for $x < -1/(2t)$.

3 Breakdown of Classical Solutions

3.1 Blow-up of the derivative and the envelope

By the proof of Theorem 2.2,

$$u_x(x,t) = \frac{u_0'(x_0)}{1 + u_0'(x_0)\,t}, \qquad x = x_0 + u_0(x_0)\,t. \tag{10}$$

The formula holds as long as the classical solution exists (Corollary 2.3, that is, for $t < -1/\inf u_0'$). If $u_0'(x_0) \geq 0$ the denominator is at least 1, and u_x remains bounded along that characteristic. If $u_0'(x_0) < 0$, the right-hand side, regarded as a formal expression along the characteristic through x_0, has a vanishing denominator at $t = -1/u_0'(x_0)$; the classical solution itself ceases to exist no later than this time, and strictly earlier unless x_0 minimises u_0' (Proposition 3.1). The characteristics

$$x(x_0,t) = x_0 + u_0(x_0)\,t, \qquad x_0 \in \mathbb{R}, \tag{11}$$

have an envelope where $\partial x/\partial x_0 = 1 + u_0'(x_0)\,t = 0$, that is, where $\Phi_t'(x_0) = 0$; this concerns the family (11), not the solution beyond its first breakdown.

3.2 The breakdown time

Proposition 3.1 (Breakdown time)**.** *Let $u_0 \in C^1(\mathbb{R})$ with $m := \inf_{x_0} u_0'(x_0) \in (-\infty, 0)$. Then the classical solution exists and is C^1 on $\mathbb{R} \times [0, T^*)$ for*

$$T^* = \frac{-1}{m}, \tag{12}$$

and $\sup_x |u_x(\cdot, t)| \to \infty$ as $t \uparrow T^*$. If the infimum is attained at $x_0 = a$ (any minimiser), then moreover no C^1 solution exists on $\mathbb{R} \times [0, T^* + \delta)$ for any $\delta > 0$, and the classical solution degenerates at time T^* at the points

$$x^* = a + u_0(a)\, T^*, \qquad t^* = T^*. \tag{13}$$

Proof. Since $r \mapsto -1/r$ is strictly increasing on $(-\infty, 0)$ and $u_0'(x_0) \geq m$ everywhere, $-1/u_0'(x_0) \geq -1/m = T^*$ for every x_0 with $u_0'(x_0) < 0$. So $\Phi_t'(x_0) > 0$ for all x_0 throughout $[0, T^*)$, and Corollary 2.3 gives a classical solution u there. Choosing x_n with $u_0'(x_n) \to m$, (10) gives $u_x \to m/(1 + mt)$ at the point with foot x_n as $n \to \infty$, so $\sup_x |u_x(\cdot, t)| \geq |m|/(1 - |m|t) \to \infty$ as $t \uparrow T^*$. Now let the infimum be attained at a, and suppose a C^1 solution v existed on $\mathbb{R} \times [0, T^* + \delta)$. By the uniqueness part of Theorem 2.2, $v = u$ on $[0, T^*)$. Along the characteristic $x(t) = a + u_0(a)t$, (10) gives $u_x = m/(1 + mt) \to -\infty$ as $t \uparrow T^*$, so v_x would be unbounded on the compact set $\{(x(t), t) : 0 \leq t \leq T^*\}$, contradicting its continuity. Substituting $x_0 = a$, $t = T^*$ into (11) gives x^*. □

If the infimum is not attained there is no first point of degeneracy, and no claim is made about extension beyond T^*.

Remark 3.2. Assume that u_0 is bounded, that the infimum m of u_0' is attained, and that $\{u_0' = m\}$ contains no interval. Then $\Phi_{T^*}' \geq 0$ vanishes only on a set with empty interior, so Φ_{T^*} is strictly increasing; it is onto $\mathbb{R}$ because u_0 is bounded, so $\Phi_{T^*}(x_0) \to \pm\infty$ as $x_0 \to \pm\infty$ (this replaces the argument of Corollary 2.3, which degenerates at $t = T^*$). Thus exactly one characteristic reaches each point at time T^*, and u, bounded by $\sup |u_0|$ by (6), develops at (x^*, T^*) only a gradient catastrophe, $|u_x| \to \infty$. If instead $u_0' \equiv m$ on an interval, then Φ_{T^*} is constant on it and characteristics carrying different values of u_0 meet at one point at time T^*, so one expects a discontinuity to form at T^*; this is not proved here (it follows from the Lax–Oleinik formula, Remark 8.5).

3.3 Example: $u_0 = -\arctan x$

Here $u_0'(x_0) = -1/(1 + x_0^2) \in [-1, 0)$ is minimised at $a = 0$ with $m = -1$, so Proposition 3.1 gives $T^* = 1$ and $x^* = 0$: the singularity first appears at $(x^*, t^*) = (0, 1)$ (Figure 1). Here $\Phi_t'(x_0) = 1 - t/(1 + x_0^2)$ first vanishes at $t = 1$, at $x_0 = 0$. At $t = 1$ the map $\Phi_1(x_0) = x_0 - \arctan x_0$ is strictly increasing but its inverse is not C^1 at 0; for $t > 1$, $\Phi_t' < 0$ on an interval around $x_0 = 0$, so Φ_t is not injective and characteristics cross.

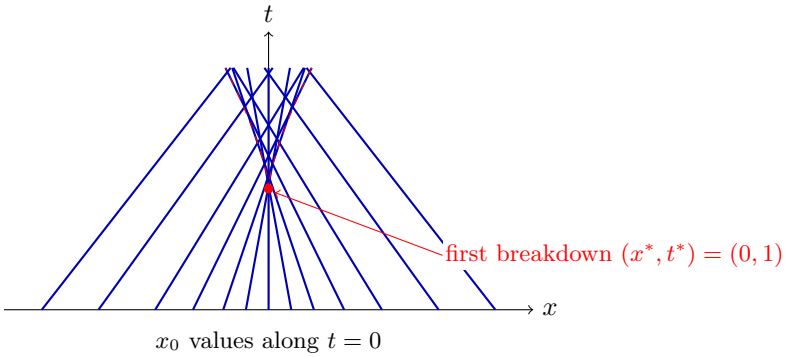


Figure 1: Characteristics $x = x_0 - \arctan(x_0)\, t$ for $u_0 = -\arctan x$, with their envelope (red, dashed), which has a cusp at the first breakdown $(x^*, t^*) = (0, 1)$. Inside the cusp ($t > 1$) three characteristics pass through each point.

By Remark 8.5 and Theorem 6.6, the entropy solution is given by the Lax–Oleinik formula (35). Here $V(y) = -\big(y \arctan y - \frac{1}{2} \ln(1 + y^2)\big)$ is even, and

$$\partial_y G(x, y, t) = \frac{\Phi_t(y) - x}{t}, \qquad \partial_y^2 G = \frac{\Phi_t'(y)}{t} = \frac{1}{t} - \frac{1}{1 + y^2}, \qquad \Phi_t(y) = y - t \arctan y.$$

$t \leq 1$. Then $\partial_y^2 G > 0$ except possibly at $y = 0$, so $G(x, \cdot, t)$ is strictly convex and its unique minimiser is the unique critical point, the characteristic foot $\Phi_t^{-1}(x)$. Thus the entropy solution is the classical solution (7).

$t > 1$. Now Φ_t is odd, with $\Phi_t' < 0$ exactly on $(-s, s)$, $s = \sqrt{t-1}$. Fix $x > 0$. Since $G(x,y,t) - G(x,-y,t) = -2xy/t$, no minimiser has $y < 0$, and $y = 0$ is not one, since $\partial_y G(x,0,t) = -x/t < 0$. On $(0, s]$ we have $\Phi_t \le 0 < x$, and on (s, ∞) Φ_t increases from a negative value to $+\infty$. Hence $G(x,\cdot,t)$ has exactly one critical point $y_+(x,t) > s$ in $(0,\infty)$, with $\Phi_t'(y_+) > 0$; it is the unique minimiser, and $u = (x - y_+)/t = -\arctan y_+$ is smooth in x by the implicit function theorem. By oddness the same holds for $x < 0$. At $x = 0$ the function $G(0,\cdot,t)$ is even with $\partial_y^2 G(0,0,t) = (1-t)/t < 0$, so 0 is not a minimiser, and the minimisers are $\pm y^*$, where $y^* > 0$ is the unique positive root of $y^* = t \arctan y^*$: a tie. Therefore, for $t > 1$, the entropy solution is smooth off $x = 0$ and has exactly one jump, stationary at $x = 0$ with

$$u_\ell = -u_r = \frac{y^*}{t} = \arctan y^* \in (0, \tfrac{\pi}{2}).$$

Moreover $y^* = \sqrt{3(1 - 1/t)}\,(1 + o(1))$ as $t \downarrow 1$ from $\arctan y = y - y^3/3 + O(y^5)$, so $u_\ell \sim \sqrt{3(t-1)}$, and $y^* > s \to \infty$ gives $u_\ell \to \pi/2$ as $t \to \infty$, matching the limits $u_0(\mp\infty) = \pm\pi/2$.

4 Weak Solutions

4.1 The integral formulation

Let $f(u) = u^2/2$. If u is a C^1 solution of $u_t + f(u)_x = 0$ with $u(\cdot,0) = u_0$ and $\varphi \in C_c^1(\mathbb{R} \times [0,\infty))$, then integration by parts in x and t gives

$$0 = \int_0^\infty \int_{\mathbb{R}} \big(u_t + f(u)_x\big)\varphi \, dx\, dt = -\int_0^\infty \int_{\mathbb{R}} \big(u\,\varphi_t + f(u)\,\varphi_x\big)\, dx\, dt - \int_{\mathbb{R}} u_0(x)\,\varphi(x,0)\, dx. \tag{14}$$

The right-hand side requires no differentiability of u.

Definition 4.1 (Weak solution)**.** Let $u_0 \in L^\infty(\mathbb{R})$. A function $u \in L^\infty(\mathbb{R} \times (0,\infty))$ is a *weak solution* of (3) (with data u_0) if

$$\int_0^\infty \int_{\mathbb{R}} \big(u\,\varphi_t + f(u)\,\varphi_x\big)\, dx\, dt + \int_{\mathbb{R}} u_0(x)\,\varphi(x,0)\, dx = 0 \tag{15}$$

for every $\varphi \in C_c^1(\mathbb{R} \times [0,\infty))$.

Reversing the integration by parts in (14) and applying the fundamental lemma of the calculus of variations shows that a bounded $u \in C^1(\mathbb{R} \times [0,\infty))$ is a weak solution with data u_0 if and only if it is a classical solution with $u(\cdot,0) = u_0$.

4.2 Piecewise smooth functions and the Rankine–Hugoniot condition

Definition 4.2 (Piecewise C^1 function)**.** Let $\sigma_1, \dots, \sigma_k : [0,\infty) \to \mathbb{R}$ be C^1 with $\sigma_1(t) < \cdots < \sigma_k(t)$ for $t > 0$ (the curves may start at a common point when $t = 0$). Put $\sigma_0 \equiv -\infty$, $\sigma_{k+1} \equiv +\infty$ and $\Omega_i = \{(x,t) :\ t > 0,\ \sigma_i(t) < x < \sigma_{i+1}(t)\}$ for $i = 0, \dots, k$. A bounded function u is *piecewise C^1 with respect to $\sigma_1, \dots, \sigma_k$* if on each Ω_i it is C^1 and u, u_x, u_t extend continuously to the closure of Ω_i in $\mathbb{R} \times (0,\infty)$, that is, to the points of the boundary curves with $t > 0$. Along $x = \sigma_i(t)$ we write $u_\ell(t)$ and $u_r(t)$ for the limits taken from Ω_{i-1} and Ω_i respectively.

Lemma 4.3 (Boundary identity)**.** *Let u be piecewise C^1 with respect to $\sigma_1, \dots, \sigma_k$ and satisfy $u_t + f(u)_x = 0$ classically in each Ω_i. Then for every $\varphi \in C_c^1(\mathbb{R} \times [0,\infty))$ and every $\varepsilon > 0$,*

$$\iint_{t>\varepsilon} \big(u\varphi_t + f(u)\varphi_x\big)\, dx\, dt = -\int_{\mathbb{R}} u(x,\varepsilon)\,\varphi(x,\varepsilon)\, dx + \sum_{i=1}^{k} \int_\varepsilon^\infty \Big[\big(f(u_\ell) - \sigma_i' u_\ell\big) - \big(f(u_r) - \sigma_i' u_r\big)\Big]\varphi(\sigma_i(t),t)\, dt. \tag{16}$$

Proof. On each $\Omega_i^\varepsilon = \Omega_i \cap \{t > \varepsilon\}$ (a Lipschitz domain, bounded after intersecting with $\operatorname{supp}\varphi$) the vector field $\varphi\,(f(u), u)$ (components in the (x,t) order) is C^1 up to the boundary, and its divergence is $u\varphi_t + f(u)\varphi_x$ because $u_t + f(u)_x = 0$ there. The divergence theorem gives

$$\iint_{\Omega_i^\varepsilon} (u\varphi_t + f(u)\varphi_x)\, dx\, dt = \oint \varphi\,\big(f(u)\,n_x + u\,n_t\big)\, d\sigma,$$

where $d\sigma\,\mathbf{n}$ equals $(1, -\sigma_{i+1}')\,dt$ on the right edge $x = \sigma_{i+1}(t)$, $(-1, \sigma_i')\,dt$ on the left edge $x = \sigma_i(t)$, and $(0,-1)\,dx$ on the bottom edge $t = \varepsilon$. Summing over i, the bottom edges give $-\int u(x,\varepsilon)\varphi(x,\varepsilon)\,dx$, and the two sides of each curve $x = \sigma_i(t)$ (from Ω_{i-1} on its left, Ω_i on its right) combine into the bracket in (16). □

Theorem 4.4 (Rankine–Hugoniot condition)**.** *Let u be piecewise C^1 with respect to $\sigma_1, \dots, \sigma_k$, and a weak solution of (3) for some data $u_0 \in L^\infty$. Then $u_t + f(u)_x = 0$ classically in each Ω_i, and along each curve $x = \sigma_i(t)$, $t > 0$,*

$$f(u_\ell(t)) - f(u_r(t)) = \sigma_i'(t)\,\big(u_\ell(t) - u_r(t)\big). \tag{17}$$

In particular, wherever $u_\ell(t) \neq u_r(t)$, and for $f(u) = u^2/2$,

$$\sigma_i'(t) = \frac{f(u_\ell) - f(u_r)}{u_\ell - u_r} = \frac{u_\ell(t) + u_r(t)}{2}. \tag{18}$$

Proof. Test functions φ supported in a single Ω_i give $\iint (u\varphi_t + f(u)\varphi_x) = 0$, so $u_t + f(u)_x = 0$ in the weak sense there; as u is C^1 on Ω_i, this holds classically, so Lemma 4.3 applies.

Now fix $t_0 > 0$, an index i, and $\varphi \in C_c^1(\mathbb{R} \times (0,\infty))$ supported in a small ball around $(\sigma_i(t_0), t_0)$ meeting no other curve. Take ε below the support of φ in (16): the left side is 0 by (15) (as $\varphi(\cdot, 0) = 0$), the bottom term vanishes, and only the curve $x = \sigma_i(t)$ contributes to the sum, so

$$0 = \int_0^\infty \Big[\big(f(u_\ell) - \sigma_i' u_\ell\big) - \big(f(u_r) - \sigma_i' u_r\big)\Big]\varphi(\sigma_i(t), t)\,dt.$$

Since $\varphi(\sigma_i(t), t)$ may be any C^1 function of t with small support, the bracket vanishes identically, which is (17). The first form of (18) follows by division, and the second from $f(u_\ell) - f(u_r) = \frac{1}{2}(u_\ell - u_r)(u_\ell + u_r)$. □

Proposition 4.5 (Piecewise smooth weak solutions)**.** *Let $u_0 \in L^\infty(\mathbb{R})$ and let u be piecewise C^1 with respect to $\sigma_1, \dots, \sigma_k$. Suppose*

(i) *$u_t + f(u)_x = 0$ classically in each Ω_i;*

(ii) *the jump relation (17) holds along each curve $x = \sigma_i(t)$, $t > 0$ (automatic where u is continuous across the curve);*

(iii) *$u(x,t) \to u_0(x)$ as $t \to 0^+$ for every $x \notin \{\sigma_1(0), \dots, \sigma_k(0)\}$.*

Then u is a weak solution of (3) with data u_0.

Proof. Fix $\varphi \in C_c^1(\mathbb{R} \times [0,\infty))$ and $\varepsilon > 0$. By (i), Lemma 4.3 applies, and by (ii) every term of the sum in (16) vanishes, so

$$\iint_{t>\varepsilon} \big(u\varphi_t + f(u)\varphi_x\big)\,dx\,dt = -\int_{\mathbb{R}} u(x,\varepsilon)\,\varphi(x,\varepsilon)\,dx.$$

Let $\varepsilon \to 0^+$. The left side tends to the integral over $t > 0$ (bounded integrand, compact support). On the right, $\varphi(\cdot,\varepsilon) \to \varphi(\cdot, 0)$ uniformly and, by (iii), $u(\cdot,\varepsilon) \to u_0$ almost everywhere with $|u| \le \|u\|_\infty$, so dominated convergence gives $-\int u_0\varphi(\cdot,0)\,dx$. This is (15). The curves need not be separated at $t = 0$, since the argument is run on $t > \varepsilon$. □

5 Non-Uniqueness of Weak Solutions

Consider (3) with the Riemann data

$$u_0(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0. \end{cases} \tag{19}$$

Characteristics emanate from $x_0 < 0$ with speed 0 and from $x_0 > 0$ with speed 1, so none covers the wedge $0 < x < t$.

Proposition 5.1. *The function*

$$u_A(x,t) = \begin{cases} 0, & x < t/2, \\ 1, & x > t/2, \end{cases} \tag{20}$$

is a weak solution of (3) with data (19).

Proof. Apply Proposition 4.5 with $k = 1$ and $\sigma_1(t) = t/2$: u_A is constant on each side, the jump moves at the Rankine–Hugoniot speed $(0+1)/2 = 1/2$ of (18), and $u_A(\cdot,t) \to u_0$ at every $x \neq 0$. □

Proposition 5.2. *The function*

$$u_B(x,t) = \begin{cases} 0, & x \le 0, \\ x/t, & 0 < x < t, \\ 1, & x \ge t, \end{cases} \tag{21}$$

is a weak solution of (3) *with data* (19).

Proof. Apply Proposition 4.5 with $k = 2$, $\sigma_1 \equiv 0$, $\sigma_2(t) = t$: u_B is constant on the outer regions, solves the PDE on $0 < x < t$ since $(u_B)_t + u_B(u_B)_x = -x/t^2 + (x/t)(1/t) = 0$, is continuous across both curves (so (17) holds trivially), and $u_B(\cdot, t) \to u_0$ at every $x \neq 0$. □

Corollary 5.3 (Non-uniqueness)**.** *The initial value problem* (3) *with data* (19) *has at least two distinct weak solutions, u_A and u_B, which differ on the wedge $0 < x < t$.*

Remark 5.4 (An infinite family). For every $\alpha \in \mathbb{R}$ let

$$u^{(\alpha)}(x,t) = \begin{cases} 0, & x < \frac{\alpha}{2}t, \\ \alpha, & \frac{\alpha}{2}t < x < \frac{1+\alpha}{2}t, \\ 1, & x > \frac{1+\alpha}{2}t. \end{cases}$$

The jumps $0 \to \alpha$ and $\alpha \to 1$ move at their Rankine–Hugoniot speeds $\alpha/2 < (1+\alpha)/2$, so Proposition 4.5 (with $k = 2$) shows that each $u^{(\alpha)}$ is a weak solution with data (19). These are distinct for distinct α, except that $u^{(0)} = u^{(1)} = u_A$.

6 The Entropy Condition

6.1 The Lax shock condition

Definition 6.1 (Lax shock condition)**.** Let u be a weak solution of a scalar conservation law $u_t + f(u)_x = 0$ with $f \in C^2$, $f'' > 0$, and suppose u has a jump discontinuity along a C^1 curve $x = \sigma(t)$ with left and right limits $u_\ell(t)$, $u_r(t)$. The jump satisfies the *Lax shock condition* if

$$f'(u_\ell(t)) > \sigma'(t) > f'(u_r(t)) \tag{22}$$

for all t.

For strictly convex f the Rankine–Hugoniot speed $(f(u_\ell) - f(u_r))/(u_\ell - u_r)$ lies strictly between $f'(u_\ell)$ and $f'(u_r)$, in the order of u_ℓ, u_r (mean value theorem), so (22) holds if and only if $u_\ell > u_r$. For $f(u) = u^2/2$ and $\sigma' = (u_\ell + u_r)/2$, (22) reduces to

$$u_\ell > \frac{u_\ell + u_r}{2} > u_r \quad \Longleftrightarrow \quad u_\ell > u_r. \tag{23}$$

Thus characteristics on both sides run into the shock. The Lax condition constrains jumps only; a derivation of $u_\ell > u_r$ from viscosity is given in Proposition 8.7.

6.2 Entropy solutions: the Oleinik condition

Definition 6.2 (Entropy solution)**.** A weak solution $u \in L^\infty$ of (3) is an *entropy solution* if

$$u(x+z,t) - u(x,t) \;\le\; \frac{z}{t} \qquad \text{for every } z > 0 \text{ and a.e. } (x,t) \in \mathbb{R} \times (0,\infty). \tag{24}$$

Condition (24) is Oleinik's one-sided Lipschitz bound [12].

Lemma 6.3 (Oleinik implies Lax)**.** *Let u be a piecewise C^1 entropy solution of Burgers' equation. Then every jump of u satisfies the Lax condition* (22)*, equivalently $u_\ell > u_r$.*

Proof. Fix a curve $x = \sigma(t)$ and $t_0 > 0$ with $u_\ell(t_0) \neq u_r(t_0)$. For $z > 0$ small enough that $[\sigma(t_0) - z, \sigma(t_0) + z]$ meets no other curve, there is a neighbourhood N of $(\sigma(t_0), t_0)$ in which every pair $(x,t), (x+z,t)$ with $x < \sigma(t) < x + z$ lies on opposite sides of the curve $x = \sigma(t)$ only, where u is C^1. Both sides of (24) are continuous on the open set $\{(x,t) \in N : x < \sigma(t) < x + z\}$, so the a.e. inequality holds at every point of it, in particular at $(\sigma(t_0) - z/2, t_0)$. Letting $z \to 0$, the left side tends to $u_r(t_0) - u_\ell(t_0)$ and the right side to 0. Hence $u_r \le u_\ell$ at t_0, and since the limits differ, $u_\ell > u_r$. □

Remark 6.4. For a piecewise C^1 solution with locally finitely many jumps, (24) holds if and only if each jump satisfies $u_\ell > u_r$ and $u_x \le 1/t$ on the smooth pieces. The Lax condition is thus the local form of (24) at jumps for piecewise C^1 solutions, whereas (24) is also meaningful for general L^∞ weak solutions, for which no jump structure is available.

Proposition 6.5. *For the Riemann data* (19) *($u_\ell = 0 < u_r = 1$), the rarefaction solution u_B of* (21) *is an entropy solution, while the shock solution u_A of* (20) *and every $u^{(\alpha)}$ of Remark 5.4 are not.*

Proof. u_B is entropy. Write $u_B(x,t) = \min\{\max\{x/t, 0\}, 1\}$, the composition of $x \mapsto x/t$ with the 1-Lipschitz function $y \mapsto \min\{\max\{y, 0\}, 1\}$. Hence $u_B(\cdot, t)$ is $(1/t)$-Lipschitz and $u_B(x+z,t) - u_B(x,t) \le z/t$, which is (24).

u_A is not. The jump $0 \to 1$ has $u_\ell < u_r$, violating (23), so by Lemma 6.3 u_A cannot be an entropy solution. Directly: for $z < t$ and every $x \in (t/2 - z, t/2)$, a set of positive measure, $u_A(x+z,t) - u_A(x,t) = 1 > z/t$.

$u^{(\alpha)}$ is not. Each $u^{(\alpha)}$ contains an upward jump: if $\alpha > 0$ the jump $0 \to \alpha$ is upward, and if $\alpha \le 0$ the jump $\alpha \to 1$ is. The same argument as for u_A applies. □

6.3 Uniqueness

Theorem 6.6 (Oleinik; [5, §3.4.3])**.** *For $u_0 \in L^\infty(\mathbb{R})$ and $f(u) = u^2/2$, the initial value problem $u_t + f(u)_x = 0$, $u(\cdot, 0) = u_0$, has at most one entropy solution in the sense of Definition 6.2, up to a set of measure zero. An entropy solution exists: the Lax–Oleinik solution [5, §3.4.2] is a weak solution satisfying* (24) *(Remark 8.5).*

For a general uniformly convex flux, Oleinik's bound reads $u(x+z,t) - u(x,t) \le C(1 + 1/t)\, z$ for some constant C, and uniqueness is proved in [5, §3.4.3] in this larger class. Since $z/t \le C(1+1/t)z$ with $C = 1$, every solution satisfying (24) lies in that class, so uniqueness there implies the statement above. For Burgers' equation the two classes coincide: any solution in the larger class equals the Lax–Oleinik solution, which satisfies (24) (Remark 8.5). The proof proceeds by duality [5]; Kruzhkov's formulation via entropy inequalities [8] yields an L^1-contraction and covers non-convex fluxes. It is not reproduced here.

7 The Riemann Problem

The entropy solution is now determined for arbitrary constant states $u_\ell, u_r \in \mathbb{R}$, that is, for the data

$$u_0(x) = \begin{cases} u_\ell, & x < 0, \\ u_r, & x > 0. \end{cases} \tag{25}$$

7.1 The rarefaction fan

Seek self-similar solutions $u(x,t) = g(\xi)$, $\xi = x/t$. Then

$$u_t + u\, u_x = -\frac{\xi}{t} g'(\xi) + \frac{g(\xi)}{t} g'(\xi) = \frac{g'(\xi)}{t}\big(g(\xi) - \xi\big) = 0, \tag{26}$$

so on each interval of ξ either $g'(\xi) = 0$ (a constant state) or $g(\xi) = \xi$ (the fan $u = x/t$). The fan connects the states $u_\ell < u_r$.

7.2 The complete solution

Theorem 7.1 (Entropy solution of the Riemann problem)**.** *The unique entropy solution of* (3) *with data* (25) *is:*

(a) *If $u_\ell > u_r$ (shock case),*

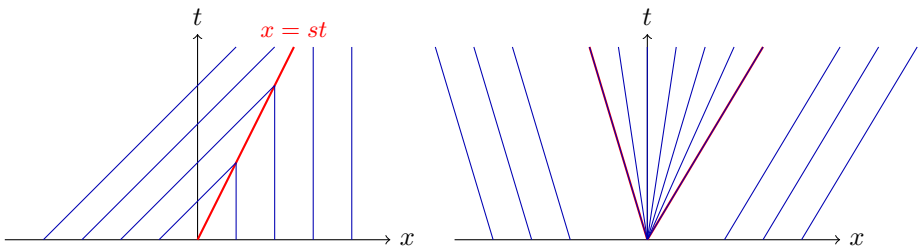
$$u(x,t) = \begin{cases} u_\ell, & x < st, \\ u_r, & x > st, \end{cases} \qquad s = \frac{u_\ell + u_r}{2}. \tag{27}$$

(b) *If $u_\ell \le u_r$ (rarefaction case),*

$$u(x,t) = \begin{cases} u_\ell, & x \le u_\ell t, \\ x/t, & u_\ell t < x < u_r t, \\ u_r, & x \ge u_r t. \end{cases} \tag{28}$$

Proof. *(a)* Apply Proposition 4.5 with $k = 1$, $\sigma_1(t) = st$: u is constant on each side (so (i) holds), the jump moves at the Rankine–Hugoniot speed $s = (u_\ell + u_r)/2$ (so (ii) holds), and for $x < 0$ (resp. $x > 0$) we have $x < st$ (resp. $x > st$) for small t, giving (iii). So u is a weak solution. It is an entropy solution: the only possible increments of $u(\cdot, t)$ are 0 and $u_r - u_\ell < 0$, so $u(x + z, t) - u(x, t) \le 0 \le z/t$, so (24) holds. Uniqueness is Theorem 6.6. The characteristics are shown in Figure 2(a).

(b) The case $u_\ell = u_r$ is the trivial constant solution, so suppose $u_\ell < u_r$. Apply Proposition 4.5 with $k = 2$, $\sigma_1(t) = u_\ell t$, $\sigma_2(t) = u_r t$ (starting at a common point). (i) u is constant on the outer regions and equals x/t on the fan $u_\ell t < x < u_r t$, where it solves the PDE by Section 7.1. (ii) u is continuous across both curves, so (17) holds trivially. (iii) For fixed $x < 0$ we have $x < u_\ell t$ for all small t, and for fixed $x > 0$, $x > u_r t$ for all small t, so $u(\cdot, t) \to u_0$ away from 0. Hence u is a weak solution. It is an entropy solution because $u(x, t) = \min\{\max\{x/t, u_\ell\}, u_r\}$ is $(1/t)$-Lipschitz in x, so (24) holds. Uniqueness is again Theorem 6.6; see Figure 2(b). □



(a) shock: $u_\ell > u_r$ (b) rarefaction: $u_\ell < u_r$

Figure 2: Characteristics for the two cases of Theorem 7.1. (a) $u_\ell = 1$, $u_r = 0$, $s = 1/2$: characteristics on both sides run into the shock $x = st$. (b) Characteristics fan out from the origin.

8 Extensions

8.1 The Cole–Hopf transformation

Theorem 8.1 (Cole–Hopf transformation [3, 7])**.** *Let $u \in C^{2,1}(\mathbb{R} \times (0, T))$ solve $u_t + u\,u_x = \nu\,u_{xx}$, with u and u_x continuous on $\mathbb{R} \times [0, T)$ (so that c below is defined), and set*

$$\hat\psi(x,t) = -\frac{1}{2\nu}\int_0^x u(y,t)\,dy, \qquad c(t) = \int_0^t \Big(\frac{u(0,\tau)^2}{4\nu} - \frac{u_x(0,\tau)}{2}\Big)d\tau, \qquad \theta(x,t) = \exp\big(\hat\psi(x,t) + c(t)\big) > 0. \tag{29}$$

Then θ solves the heat equation $\theta_t = \nu\,\theta_{xx}$, and $u = -2\nu\,\theta_x/\theta$.

Proof. Since c depends on t only, $\theta_x/\theta = \hat\psi_x = -u/(2\nu)$, which gives $u = -2\nu\theta_x/\theta$. For the heat equation, let $w = \hat\psi_t - \nu\hat\psi_{xx} - \nu\hat\psi_x^2$. Using $\hat\psi_x = -u/(2\nu)$, $\hat\psi_{xx} = -u_x/(2\nu)$, $\hat\psi_{xt} = -u_t/(2\nu)$ and $\hat\psi_{xxx} = -u_{xx}/(2\nu)$,

$$w_x = \hat\psi_{xt} - \nu\hat\psi_{xxx} - 2\nu\hat\psi_x\hat\psi_{xx} = -\frac{1}{2\nu}\big(u_t + u\,u_x - \nu\,u_{xx}\big) = 0. \tag{30}$$

Hence w depends on t only and may be evaluated at $x = 0$, where $\hat\psi(0, t) = 0$ gives $\hat\psi_t(0, t) = 0$:

$$w(t) = w(0,t) = \frac{u_x(0,t)}{2} - \frac{u(0,t)^2}{4\nu} =: C(t). \tag{31}$$

With $\psi = \hat\psi + c$ we have $c' = -C$, so $\psi_t - \nu\psi_{xx} - \nu\psi_x^2 = C + c' = 0$. Since $\theta = e^\psi$ satisfies $\theta_t = \theta\psi_t$ and $\theta_{xx} = \theta(\psi_{xx} + \psi_x^2)$, this is $\theta_t = \nu\theta_{xx}$. □

Remark 8.2. The factor $e^{c(t)}$ is independent of x, so it affects neither $u = -2\nu\theta_x/\theta$ nor the initial data ($c(0) = 0$). It cannot be omitted: $e^{\hat\psi}$ satisfies $\partial_t\theta = \nu\theta_{xx} + C(t)\theta$, which is the heat equation only if $C \equiv 0$.

8.2 Exact solution via the heat kernel

By Theorem 8.1, a classical solution of (1) that is bounded on $\mathbb{R}\times[0,T']$ for each $T'<T$ (this is what makes θ obey the growth bound below) corresponds to a solution of the heat equation with initial data

$$\theta_0(x)=\exp\left(-\frac{1}{2\nu}\int_0^x u_0(y)\,dy\right)\le e^{\|u_0\|_\infty|x|/(2\nu)}. \tag{32}$$

The Cauchy problem for the heat equation has a unique solution in the class of at most $e^{a|x|^2}$ growth [5, §2.3], which contains this data; it is given by

$$\theta(x,t)=\int_{\mathbb{R}}\frac{1}{\sqrt{4\pi\nu t}}\,\exp\left(-\frac{(x-y)^2}{4\nu t}\right)\theta_0(y)\,dy. \tag{33}$$

Substituting (32)–(33) into $u=-2\nu\,\theta_x/\theta$ and differentiating under the integral gives the *Cole–Hopf formula*

$$u(x,t)=\frac{\displaystyle\int_{\mathbb{R}}\frac{x-y}{t}\,\exp\left(-\frac{G(x,y,t)}{2\nu}\right)dy}{\displaystyle\int_{\mathbb{R}}\exp\left(-\frac{G(x,y,t)}{2\nu}\right)dy},\qquad G(x,y,t)=\int_0^y u_0(r)\,dr+\frac{(x-y)^2}{2t}. \tag{34}$$

Corollary 8.3 (Global smoothness; Hopf [7])**.** *For every bounded measurable u_0 and every $\nu>0$, formula* (34) *defines a C^∞ solution $u^{(\nu)}$ of* (1) *on $\mathbb{R}\times(0,\infty)$. Moreover $\|u^{(\nu)}\|_\infty\le\|u_0\|_\infty$, $u^{(\nu)}(\cdot,t)\to u_0$ in L^1_{loc} and at every continuity point of u_0 as $t\to0^+$, and $u^{(\nu)}$ is the unique bounded solution with these properties.*

For $t>0$, θ in (33) is smooth and positive, and for any such solution θ of the heat equation $u=-2\nu\theta_x/\theta$ solves (1): with $\psi=\ln\theta$, $\psi_t-\nu\psi_{xx}-\nu\psi_x^2=0$, and differentiating in x gives $(-u/2\nu)_t-\nu(-u/2\nu)_{xx}-\nu\big((u/2\nu)^2\big)_x=0$, i.e. $u_t+u\,u_x=\nu u_{xx}$. This proves the first assertion for merely measurable u_0 (Theorem 8.1 is not needed in this direction). The bound, the sense in which the data are attained, and uniqueness are due to Hopf [7] and are cited without proof.

8.3 The vanishing viscosity limit

Theorem 8.4 (Vanishing viscosity limit [7])**.** *Let $u_0\in L^\infty(\mathbb{R})$, let $u^{(\nu)}$ be the solution of* (1) *with data u_0 given by* (34)*, and let u be the entropy solution of* (2) *with the same data (Definition 6.2; for Riemann data, Theorem 7.1). Then $u^{(\nu)}\to u$ pointwise almost everywhere and in L^1_{loc} as $\nu\to0^+$.*

The proof is cited, not given. Hopf [7], who predates Oleinik's condition, proves by Laplace's method applied to (34) that $u^{(\nu)}$ converges to the Lax–Oleinik solution (35); Remark 8.5 below indicates why this solution satisfies (24), and Theorem 6.6 identifies it with u.

Remark 8.5 (The Lax–Oleinik formula). As $\nu\to0$ the weight $\exp(-G/2\nu)$ in (34) concentrates at the minimisers of $y\mapsto G(x,y,t)$. Where the minimiser $y^*(x,t)$ is unique, the limit is

$$u(x,t)=\frac{x-y^*(x,t)}{t},\qquad y^*(x,t)=\arg\min_y\left\{V(y)+\frac{(x-y)^2}{2t}\right\},\quad V(y)=\int_0^y u_0, \tag{35}$$

the *Lax–Oleinik formula* [9, 12], [5, §3.4.2]; a jump occurs where two minimisers tie. Since $x\mapsto y^*(x,t)$ is nondecreasing [5, §3.4.2], for $z>0$

$$u(x+z,t)-u(x,t)=\frac{z-\big(y^*(x+z,t)-y^*(x,t)\big)}{t}\;\le\;\frac{z}{t},$$

which is (24). As (35) also defines a weak solution [5, §3.4.2], the Lax–Oleinik solution is an entropy solution in the sense of Definition 6.2. The same bound shows $u(x+,t)-u(x-,t)\le0$ wherever one-sided limits exist: every jump is downward. The minimiser is unique for almost every (x,t) [5, §3.4.2], so (35) defines u a.e.

Remark 8.6 (The Riemann problem from the Lax–Oleinik formula). For the data (25), $V(y)=u_\ell y$ for $y<0$ and $V(y)=u_r y$ for $y>0$, so $G(y)=V(y)+(x-y)^2/2t$ is convex on each half-line, with critical points $y_\ell=x-u_\ell t$ (admissible iff $x\le u_\ell t$) and $y_r=x-u_r t$ (iff $x\ge u_r t$), and a kink at $y=0$ with one-sided derivatives $u_\ell-x/t$ and u_r-x/t. If $u_\ell<u_r$ and $u_\ell t<x<u_r t$, only the kink is a local minimiser and $u=x/t$; outside this range the single admissible critical point gives u_ℓ or u_r. This is (28). If $u_\ell>u_r$ and $u_r t\le x\le u_\ell t$, both y_ℓ and y_r are admissible and the kink is a local maximum, with $G(y_\ell)=u_\ell x-u_\ell^2 t/2$ and $G(y_r)=u_r x-u_r^2 t/2$, so $G(y_\ell)-G(y_r)=(u_\ell-u_r)\big(x-\frac{u_\ell+u_r}{2}t\big)$. The tie, hence the jump, is at $x=\frac{u_\ell+u_r}{2}t$, recovering (27) and the Rankine–Hugoniot speed.

8.3.1 Viscous shock profiles

Proposition 8.7 (Viscous shock profiles)**.** *Let $u_\ell \neq u_r$ and $\nu > 0$. Equation* (1) *has a travelling-wave solution $u(x,t) = \phi(x - st)$, with $\phi \in C^2$ bounded, $\phi(\eta) \to u_\ell$ as $\eta \to -\infty$, $\phi(\eta) \to u_r$ as $\eta \to +\infty$ and $\phi' \to 0$ as $\eta \to \pm\infty$, if and only if $u_\ell > u_r$. In that case $s = (u_\ell + u_r)/2$ and, with $\Delta = u_\ell - u_r$, ϕ is unique up to translation $\eta \mapsto \eta + \eta_0$; the translate with $\phi(0) = s$ is*

$$\phi(\eta) = s - \frac{\Delta}{2}\tanh\Big(\frac{\Delta\,\eta}{4\nu}\Big). \tag{36}$$

Moreover $\|\phi - (s - \frac{\Delta}{2}\operatorname{sgn}\eta)\|_{L^1(\mathbb{R})} = 4\ln 2\,\nu$, independent of s and Δ.

Proof. Substituting $u = \phi(x - st)$ gives $-s\phi' + \phi\phi' = \nu\phi''$, and integrating once from $-\infty$ (where $\phi' \to 0$),

$$\nu\phi' = \tfrac{1}{2}\phi^2 - s\phi - K, \qquad K = \tfrac{1}{2}u_\ell^2 - su_\ell.$$

Letting $\eta \to +\infty$ shows that the right-hand side must also vanish at u_r, so u_ℓ, u_r are the two roots of the quadratic: $s = (u_\ell + u_r)/2$ (the Rankine–Hugoniot speed) and

$$\nu\phi' = \tfrac{1}{2}(\phi - u_\ell)(\phi - u_r). \tag{37}$$

This autonomous first-order ODE cannot be crossed at its equilibria u_ℓ, u_r, so a connecting ϕ lies strictly between them. On $(\min(u_\ell,u_r), \max(u_\ell,u_r))$ the right side of (37) is negative, so ϕ is decreasing. If $u_\ell < u_r$ it would have to increase from u_ℓ to u_r, a contradiction; if $u_\ell > u_r$, (37) is solved by (36); any other solution is a translate, by uniqueness for the ODE. For the L^1 distance, by oddness and $\int_0^\infty (1 - \tanh\zeta)\,d\zeta = \ln 2$,

$$\int_{\mathbb{R}} \tfrac{\Delta}{2}\Big|\tanh\tfrac{\Delta\eta}{4\nu} - \operatorname{sgn}\eta\Big|\,d\eta = \Delta\int_0^\infty \Big(1 - \tanh\tfrac{\Delta\eta}{4\nu}\Big)d\eta = \Delta\cdot\frac{4\nu}{\Delta}\ln 2 = 4\ln 2\,\nu.$$

□

Hence a jump is the limit of viscous travelling waves, of thickness $O(\nu/\Delta)$, exactly when $u_\ell > u_r$. This motivates the Lax condition (23) from vanishing viscosity. That u_A of Section 5, with its upward jump, is not the vanishing-viscosity limit is the content of Theorem 8.4 together with Proposition 6.5.

For Riemann data the Cole–Hopf solution is available in closed form since G is piecewise quadratic (`code/cole_hopf_riemann.py`). For the shock $(1,0)$ its $L^1(-2,2)$ distance to the entropy solution at $t = 1$ equals $2.773\,\nu$ for $\nu \le 10^{-3}$, the layer value $4\ln 2\,\nu$ of Proposition 8.7; for the rarefaction $(0,1)$ the error divided by ν grows by about 4.4 per decade of ν (observed, not derived).

8.4 Godunov's method

8.4.1 The scheme

Let $\mathbb{R}$ be divided into cells $[x_{j-1/2}, x_{j+1/2}]$ of width Δx, with $t^n = n\Delta t$ and cell averages $\bar u_j^n = \Delta x^{-1}\int_{x_{j-1/2}}^{x_{j+1/2}} u(x,t^n)\,dx$ approximated by U_j^n. Integrating $u_t + f(u)_x = 0$ over a cell and a time step gives

$$\bar u_j^{n+1} = \bar u_j^n - \frac{\Delta t}{\Delta x}\left(F_{j+1/2}^n - F_{j-1/2}^n\right), \qquad F_{j+1/2}^n = \frac{1}{\Delta t}\int_{t^n}^{t^{n+1}} f\big(u(x_{j+1/2},t)\big)\,dt, \tag{38}$$

for the exact cell averages $\bar u_j^n$. The scheme replaces $F_{j+1/2}^n$ by a numerical flux $\widehat f_{j+1/2}$ depending on U_j^n, U_{j+1}^n, giving $U_j^{n+1} = U_j^n - \frac{\Delta t}{\Delta x}(\widehat f_{j+1/2} - \widehat f_{j-1/2})$. Godunov's method [6] takes $\widehat f_{j+1/2}$ to be the flux at $x/t = 0$ of the exact Riemann solution with data (U_j^n, U_{j+1}^n) (Theorem 7.1):

$$\widehat f_{j+1/2} = f\big(\omega(0;\,U_j^n, U_{j+1}^n)\big), \tag{39}$$

where $\omega(\xi;u_\ell,u_r)$ is the entropy solution with data (u_ℓ,u_r) at $\xi = x/t$. By (27)–(28),

$$\omega(0;u_\ell,u_r) = \begin{cases} u_\ell, & u_\ell > u_r,\ \frac{u_\ell+u_r}{2} > 0,\\ u_r, & u_\ell > u_r,\ \frac{u_\ell+u_r}{2} < 0,\\ u_\ell, & u_\ell \le u_r,\ u_\ell \ge 0,\\ u_r, & u_\ell \le u_r,\ u_r \le 0,\\ 0, & u_\ell < 0 < u_r, \end{cases} \tag{40}$$

the last case corresponding to $\xi = 0$ lying inside the fan. If $u_\ell = -u_r$ the value $\omega(0)$ is ambiguous, but $f(u_\ell) = f(u_r)$, so (39) is unaffected.

8.4.2 Stability and convergence

The scheme is stable and monotone under the CFL condition

$$\Delta t \le \frac{\Delta x}{\max_j |U_j^n|}, \tag{41}$$

which ensures that waves from neighbouring interfaces do not reach $x_{j+1/2}$ within one step [10, Ch. 4].

Theorem 8.8 ([4, Theorem 1 and §1]). *Let $u_0 \in L^1(\mathbb{R}) \cap L^\infty(\mathbb{R})$ with $a \le u_0 \le b$ a.e., and let $\lambda = \Delta t/\Delta x$ be fixed with $\lambda \max_{a\le u\le b} |f'(u)| \le 1$. Let u^Δ be the piecewise constant function equal to U_j^n on $[x_{j-1/2}, x_{j+1/2}) \times [t^n, t^{n+1})$, where U_j^0 is the cell average of u_0 and the U_j^n are the Godunov approximations. Then for every $T > 0$*

$$\lim_{\Delta t\to 0} \sup_{0\le t\le T} \int_{\mathbb{R}} \left|u^\Delta(x,t) - u(x,t)\right| dx = 0,$$

where u is the Kružkov entropy solution [8] with data u_0.

Crandall and Majda prove this for any consistent conservation-form scheme that is monotone on $[a, b]$ with Lipschitz continuous numerical flux, and verify that Godunov's scheme for $f'' > 0$ is of this type. The theorem asserts convergence, not a rate. For Burgers' equation the Kružkov solution coincides with the Oleinik solution of Theorem 6.6 (standard; see [5, §3.4] and [8]). The Riemann data used below are not in $L^1(\mathbb{R})$; they are handled by localisation. Replace them by 0 outside $[-5, 5]$, which keeps u_0 within $[0, 1]$. As $|u| \le 1$, the exact solution on $[-2, 2] \times [0, 1]$ is unchanged, and at $\lambda = 0.45$ the numerical domain of dependence over $t \le 1$ has half-width at most $t/\lambda \le 1/0.45 < 2.23$, so the dependence interval of $[-2, 2]$ lies in $[-4.23, 4.23] \subset [-5, 5]$ and the computed solution there is unchanged too. In the experiments $\max_j |U_j^n| = 1$ throughout, so the adaptive step $\Delta t = 0.45\,\Delta x/\max_j |U_j^n|$ gives the constant $\lambda = 0.45$, satisfying the hypothesis on λ for data in $[0, 1]$ (the final step, shortened to land on $t = 1$, has smaller λ).

8.4.3 Numerical results

The scheme was implemented (`code/godunov.py`) on $x \in [-2, 2]$ with N cells, cell averages of the Riemann data (the discontinuity lies on a cell interface for even N; for odd N the cell containing 0 holds a mixed average), outflow boundary conditions and $\Delta t = 0.45\,\Delta x/\max_j |U_j^n|$ (the last step is shortened so that the final time is exactly $t = 1$), for the rarefaction data $(u_\ell, u_r) = (0, 1)$ and the shock data $(1, 0)$. Errors are measured in $L^1(-2, 2)$ against the exact solution of Theorem 7.1, integrated cell by cell. The case-based flux (39) agrees with the closed form $\min_{[u_\ell,u_r]} f$ (if $u_\ell \le u_r$) and $\max_{[u_r,u_\ell]} f$ (otherwise) on 10^5 random state pairs, and total mass equals the initial mass plus the net boundary flux to ten digits. The outflow condition has a negligible effect: for $t \le 1$ the exact solution is constant near $x = \pm 2$, and repeating $N = 400$ on $[-4, 4]$ with the same Δx changes the computed values on $[-2, 2]$ by at most 3×10^{-14}. Figure 3 shows $N = 400$: the computed shock lies at $x = st = 0.5$ and the fan has slope $1/t$, with no oscillation.

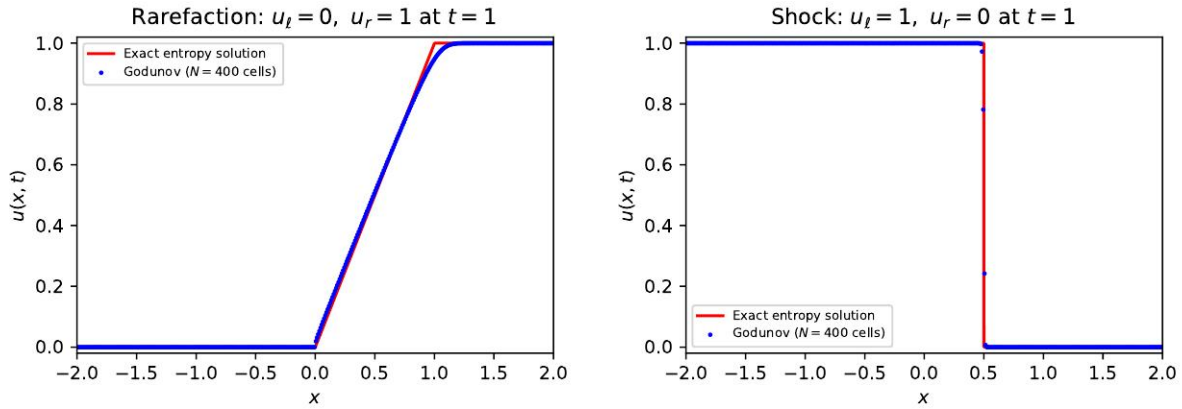


Figure 3: Godunov scheme (dots, $N = 400$) against the exact entropy solution (solid) at $t = 1$. Left: rarefaction data $(0, 1)$. Right: shock data $(1, 0)$.

N	Δx	L^1 (rarefaction)	order	L^1 (shock)	order
200	2.0×10^{-2}	3.05×10^{-2}	—	9.91×10^{-3}	—
400	1.0×10^{-2}	1.82×10^{-2}	0.74	4.98×10^{-3}	0.99
800	5.0×10^{-3}	1.07×10^{-2}	0.77	2.50×10^{-3}	0.99
1600	2.5×10^{-3}	6.14×10^{-3}	0.80	1.24×10^{-3}	1.01
3200	1.25×10^{-3}	3.48×10^{-3}	0.82	6.23×10^{-4}	0.99
6400	6.25×10^{-4}	1.95×10^{-3}	0.84	3.13×10^{-4}	0.99

Table 1: L^1 errors at $t = 1$ and observed orders $\log_2(e_{N/2}/e_N)$. All N are multiples of 8, so the shock lies on a cell interface.

Shock. The shock $x = st = 0.5$ lies on a cell interface iff $8 \mid N$. For such N the error is $0.50\,\Delta x$ to two digits (0.495–0.501) and the observed order is ≈ 1 (Table 1). For $N = 97, \dots, 103$ the error ranges from $0.55\,\Delta x$ to $0.66\,\Delta x$ (e.g. $0.655\,\Delta x$ at $N = 100$, against $0.50\,\Delta x$ at $N = 96, 104$), so orders computed between misaligned and aligned resolutions are unreliable ($N = 100 \to 200$ would give 1.40); the table therefore uses $N = 200 \cdot 2^k$. In all cases the error is proportional to Δx.

Rarefaction. The error decreases monotonically with observed order $0.74, 0.77, 0.80, 0.82, 0.84$, still increasing. A first-order rate is therefore not established by these experiments. We conjecture that the excess error comes from the two corners of the fan, where the exact solution is only Lipschitz; no experiment here isolates the corners.

8.4.4 Numerical diffusion as vanishing viscosity

For the shock data the scheme is monotone, so by the discrete maximum principle all states remain in $[0, 1]$. All characteristic speeds are then nonnegative, so the Godunov flux is the upwind flux $f(U_j^n)$, and the scheme reads $U_j^{n+1} = U_j^n - \lambda(f(U_j^n) - f(U_{j-1}^n))$, $\lambda = \Delta t/\Delta x$. Taylor expansion gives the modified equation

$$u_t + u\,u_x = \frac{\Delta x}{2}\big(u(1 - \lambda u)\,u_x\big)_x + O(\Delta x^2), \tag{42}$$

a Burgers-type equation with state-dependent viscosity $\nu(u) = \frac{\Delta x}{2} u(1 - \lambda u)$, nonnegative exactly under (41). Its travelling shock $1 \to 0$ of speed $\frac{1}{2}$ satisfies $\nu(u)u_\eta = \frac{1}{2}u(u - 1)$ in $\eta = x - t/2$; in units of Δx,

$$\frac{du}{d\eta} = \frac{u - 1}{1 - \lambda u}, \qquad \eta(u) = \eta_0 + (1 - \lambda)\ln(1 - u) - \lambda u. \tag{43}$$

The profile decays exponentially to 1 as $\eta \to -\infty$ and reaches 0 at a finite η_0, which conservation of mass fixes as $\eta_0 = 1 - \lambda/2$. The L^1 distance from the sharp shock is then

$$\frac{\|u - u_{\text{exact}}\|_{L^1}}{\Delta x} = 2\int_0^{u_0} \frac{u(1 - \lambda u)}{1 - u}\,du, \qquad (1 - \lambda)\ln(1 - u_0) - \lambda u_0 = -\Big(1 - \frac{\lambda}{2}\Big). \tag{44}$$

Table 2 compares this with the measured shock error on aligned grids, where λ equals the CFL number since $\max|U| = 1$ (`code/numerical_viscosity.py`).

CFL $= \lambda$	measured ($N = 1600$)	predicted	(pred.−meas.)/meas.
0.10	0.627	0.686	+9.4%
0.20	0.589	0.635	+7.8%
0.30	0.553	0.586	+6.0%
0.45	0.496	0.511	+3.0%
0.60	0.448	0.438	−2.2%
0.80	0.352	0.342	−2.8%
0.95	0.300	0.272	−9.3%

Table 2: Shock L^1 error divided by Δx: measured (grids with $8 \mid N$) against the modified-equation prediction; the last column is computed from the tabulated values.

The prediction is within about 3% at CFL 0.45 (+3.0%) and within 10% throughout. The value at CFL 0.95 is not yet asymptotic (0.334 at $N = 400$, 0.300 at $N = 1600$). Equating the error with the viscous-shock value $4\ln 2\,\nu_{\text{eff}}$ of Proposition 8.7 ($s = \frac{1}{2}$, $\Delta = 1$) defines an effective viscosity: the measured $0.50\,\Delta x$ gives $\nu_{\text{eff}} \approx 0.180\,\Delta x$, the modified-equation profile gives $0.184\,\Delta x$, and $\nu(\frac{1}{2}) = \frac{\Delta x}{2} \cdot \frac{1}{2}(1 - \frac{\lambda}{2})$ gives $0.194\,\Delta x$ at $\lambda = 0.45$. Since $\nu_{\text{eff}} = O(\Delta x) \to 0$, the scheme acts in the shock layer as a vanishing viscosity, in accordance with Theorem 8.4. This is a consistency check rather than a proof: the numerical shock spans a few cells and (42) omits the $O(\Delta x^2)$ terms.

8.4.5 A consistent scheme that selects the wrong solution

Consider the flux $\widehat{f}_{j+1/2} = f(U_j)$ if $\frac{1}{2}(U_j + U_{j+1}) > 0$ and $f(U_{j+1})$ otherwise. It is consistent and conservative, but not monotone. For the Riemann data $U_j = -1$ ($j \le 0$), $U_j = +1$ ($j > 0$) every numerical flux equals $f(\pm 1) = \frac{1}{2}$, so $U^{n+1} = U^n$ exactly: the scheme returns the stationary upward jump, the analogue for these data of u_A in Section 5, which is a weak solution by (18) with speed 0 but not an entropy solution. Godunov's flux at the same interface is $f(0) = 0$, and the scheme produces the fan $u = x/t$ of Theorem 7.1. At $t = 1$ the $L^1(-2, 2)$ distance to the entropy solution is 1.000 for the non-monotone scheme for all $N = 200, 400, 800$, since $\int |\operatorname{sgn} x - \operatorname{clip}(x, -1, 1)|\, dx = 1$, while for Godunov's scheme it is 6.1×10^{-2}, 3.6×10^{-2}, 2.1×10^{-2}. By the Lax–Wendroff theorem [10, Ch. 12] limits of consistent conservative schemes are weak solutions, so consistency and conservation alone do not select the entropy solution: monotonicity (Theorem 8.8) does.

9 Conclusion

The weak formulation extends solutions of (2) beyond breakdown but does not determine them: already for Riemann data there are infinitely many weak solutions (Section 5). The Oleinik condition, whose local form at a jump is the Lax condition $u_\ell > u_r$, restores uniqueness (Sections 6–7). It is the physically correct selection because it is the one made by vanishing viscosity: the Cole–Hopf solutions converge to the entropy solution (Theorem 8.4, Remark 8.5), and a viscous travelling wave connecting u_ℓ to u_r exists if and only if $u_\ell > u_r$ (Proposition 8.7). The underlying theory is standard [5, §3.4].

The Godunov scheme of Section 8.4 was tested on Riemann data. The shock error is $0.50\,\Delta x$ on grids aligned with the shock and 0.55–$0.66\,\Delta x$ on the misaligned grids tested ($N = 97, \dots, 103$); the rarefaction error has observed order ≈ 0.8, still increasing. The modified equation (42) predicts the shock error to within about 3% at CFL 0.45 and within about 10% for CFL numbers in $[0.1, 0.95]$, corresponding to an effective viscosity $\nu_{\text{eff}} \approx 0.18\,\Delta x$. This is a heuristic consistency check; the rigorous result is Theorem 8.8 [4].

The proofs of Theorems 6.6, 8.4 and 8.8 are cited, not reproduced, and the long-time N-wave asymptotics for compactly supported data [9], [5, §3.4.5] are not treated.

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